Breaking this into two parts, we see that the first sum telescopes to give $\mathrm{W}(\tau)^2$ (as $\mathrm{W}(0) = 0$). On the other hand, from (6.7), we have

$$\lim_{\nu\to\infty}\mathbb{E}\sum_{k=0}^{\nu-1}\Big[(\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t))^2\Big] = \tau,$$

but an even stronger statement is possible. Since we are summing a large collection of random variables $\Delta\mathrm{W}_k$ whose means are δt, the Central Limit Theorem tells us that, if we assume it to be well defined, the variance of the sum of squares from τ must tend to zero as a function of ν. Thus we may understand the convergence in a somewhat stronger sense than convergence in expectation. We make this precise in the following proposition

Proposition 6.2 *Let $\nu\delta t = \tau$ then*

$$\text{l.i.m.}_{\nu\to\infty}\sum_{k=0}^{\nu-1}\Big[(\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t))^2\Big] = \tau,$$

where, for a sequence of random variables, A_ν *and a random variable* B*, we have*

$$\text{l.i.m.}_{\nu\to\infty}\mathrm{A}_\nu = \mathrm{B} \Leftrightarrow \lim_{\nu\to\infty}\mathbb{E}(\mathrm{A}_\nu - \mathrm{B})^2 = 0.$$

(This is referred to as convergence in the mean square sense.)

Proof With τ, ν fixed, we have $\delta t = \tau/\nu$.

Let $\Delta\mathrm{W}_{\nu,k} = \mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t) = \mathrm{W}((k+1)\tau/\nu) - \mathrm{W}(k\tau/\nu)$ and observe that

$$\sum_{k=0}^{\nu-1}\Delta\mathrm{W}_{\nu,k}^2 - \tau = \sum_{k=0}^{\nu-1}\Big[\Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}\Big].$$

Then

$$I_\nu \overset{\text{def}}{=} \mathbb{E}\left(\sum_{k=0}^{\nu-1}\Delta\mathrm{W}_{\nu,k}^2 - \tau\right)^2 = \mathbb{E}\left(\sum_{k=0}^{\nu-1}\Big[\Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}\Big]\right)^2.$$

Terms of the form $Z_{\nu,k} \overset{\text{def}}{=} \Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}$ are independent and have mean zero. Therefore

$$\mathbb{E}\left(\sum_{k=0}^{\nu-1}Z_{\nu,k}\right)^2 = \mathbb{E}\sum_{k=0}^{\nu-1}Z_{\nu,k}^2 - 2\mathbb{E}\sum_{k\neq l}Z_{\nu,k}Z_{\nu,l} = \mathbb{E}\sum_{k=0}^{\nu-1}Z_{\nu,k}^2,$$

where the last equation follows from independence of the increments ($\mathbb{E}Z_k Z_l = 0$ if $k \neq l$).

Also observe that $\Delta \mathrm{W}_k$ is a normally distributed random variable with the following properties

$$\mathbb{E}\Delta \mathrm{W}_k = 0, \qquad \mathbb{E}\Delta \mathrm{W}_k^2 = \delta t = \tau/\nu, \qquad \mathbb{E}\Delta \mathrm{W}_k^4 = 3\delta t^2 = 3(\tau/\nu)^2.$$

Thus we have

$$\begin{aligned} I_\nu &= \mathbb{E}\sum_{k=0}^{\nu-1}(\Delta \mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu})^2 \\ &= \sum_{k=0}^{\nu-1}\mathbb{E}\left(\Delta \mathrm{W}_{\nu,k}^4 - 2\frac{\tau}{\nu}\Delta \mathrm{W}_{\nu,k}^2 + \frac{\tau^2}{\nu^2}\right) \\ &= \sum_{k=0}^{\nu-1}\left(3\left(\frac{\tau}{\nu}\right)^2 - 2\left(\frac{\tau}{\nu}\right)^2 + \frac{\tau^2}{\nu^2}\right) = 2\frac{\tau^2}{\nu^2}\nu = 2\frac{\tau^2}{\nu}. \end{aligned}$$

Thus $I_\nu \to 0$ as $\nu \to 0$, which is the desired result. □

The calculation implies that

$$\int_0^\tau \mathrm{W}(t)\mathrm{dW}(t) = \frac{1}{2}\left(\mathrm{W}(\tau)^2 - \tau\right).$$

In contrast, the Stratonovich (midpoint) integral (denoted by $\int_0^\tau \circ \mathrm{dW}(t)$) can be calculated as

$$\int_0^\tau \mathrm{W}(t) \circ \mathrm{dW}(t) = \mathrm{l.i.m.}_{\nu\to\infty}\sum_{k=0}^{\nu-1}\mathrm{W}(k\delta t + \delta t/2)[\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)].$$

Writing

$$\mathrm{W}(k\delta t + \delta t/2) = \mathrm{W}\left(\frac{(k+1)\delta t + k\delta t}{2}\right) = \frac{1}{2}\left(\mathrm{W}((k+1)\delta t) + \mathrm{W}(k\delta t)\right) + \mathcal{O}(\sqrt{\delta t}),$$

allows us to evaluate the integral as

$$\int_0^\tau \mathrm{W}(t) \circ \mathrm{dW}(t) = \frac{1}{2}\mathrm{l.i.m.}_{\nu\to\infty}\sum_{k=0}^{\nu-1}\left[\mathrm{W}((k+1)\delta t)^2 - \mathrm{W}(k\delta t)^2\right] = \frac{1}{2}\mathrm{W}(\tau)^2.$$